Che 312: Homework Assignment #6

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Problems: 5.6, 5.70, 6.3, 6.34

91.5/100

$L_c = \frac{V}{A_s}$ $Bi = \frac{h L_c}{k}$

Problem 5.6

$L_c = ?$ $Bi = ?$ is lumped cap. approx valid?

(a)

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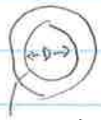


$A_c = 5\,mm^2$

$D = 50\,mm$

$k = 2.3\ W/mK$

$h = 50\ W/m^2K$

$V = (A_c)(D)(\pi)$

$A_s = 2\pi r_o \pi D$

$r_o = \sqrt{\frac{A_c}{\pi}}$

$$L_c = \frac{A_c \pi \cancel{D}}{2\pi (A_c \pi)^{-1/2} \cancel{\pi D}} = \frac{1}{2}\sqrt{\frac{A_c}{\pi}} = \frac{1}{2}\sqrt{\frac{5mm^2}{\pi}}$$

$L_c = 0.631\,mm = 0.631\times10^{-3}\,m$

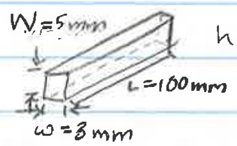
Lc_2?

Bi_2?

$$Bi = \frac{(50\ W/m^2K)(0.631\times10^{-3}m)}{2.3\ W/mK} = 0.0137$$

$$\boxed{L_c = 0.631\times10^{-3}m \quad Bi = 0.0137}$$ The lumped cap. approx is valid

(b)



$h = 15\ W/m^2K$

Table A-1

$k = 14.9\ W/mK$

$V = (L)(W)(w) = (100mm)(5mm)(3mm) = 1500\,mm^3$

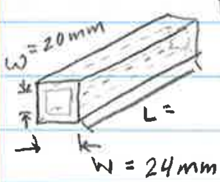
$A_s = 2(wL + WL + wW) = 2(3\cdot100 + 5\cdot100 + 3\cdot5) = 1630\,mm^2$

$$L_c = \frac{1500\,mm^3}{1630\,mm^2} = 0.920\,mm = 0.920\times10^{-3}\,m$$

$$Bi = \frac{(15\ W/m^2K)(0.920\times10^{-3}m)}{(14.9\ W/mK)} = 9.26\times10^{-4}$$

$$\boxed{L_c = 0.920\times10^{-3}m \quad Bi = 9.26\times10^{-4}}$$ The lumped cap. approx is valid.

(c)



$k = 177\ \frac{W}{mK}$

$h = 37\ W/m^2K$

$V = (W^2 - w^2)L$

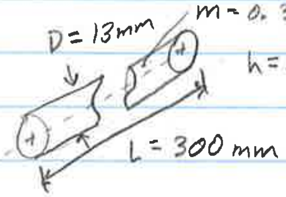
$A_s = 4WL$

$$L_c = \frac{(W^2-w^2)\cancel{L}}{4W\cancel{L}} = \frac{(24^2 - 20^2)}{(4)(24)} = 1.83\,mm = 1.83\times10^{-3}\,m$$

$$Bi = \frac{(37\ W/m^2K)(1.83\times10^{-3}m)}{177\ W/mK} = 3.83\times10^{-4}$$

$$\boxed{L_c = 1.83\times10^{-3}m \quad Bi = 3.83\times10^{-4}}$$ Lumped cap. approx is valid

(d)



$m = 0.328\,kg$

$h = 30\ W/m^2K$

$\rho = \frac{m}{V}$

$$V = \frac{\pi D^2 L}{4} = \frac{\pi(0.013m)^2(0.3m)}{4} = 3.98\times10^{-5}\,m^3$$

$$\rho = \frac{0.328\,kg}{3.98\times10^{-5}m^3} = 8241.2\ kg/m^3 \Leftarrow \text{AISI 316}$$

$k = 13.4\ \frac{W}{mK}$

$$A_s = \pi D L + \frac{2\pi D^2}{4} = \pi(0.013m)(0.3m) + \frac{2\pi(0.013m)^2}{4} = 0.0125\,m^2$$

$$L_c = \frac{3.98\times10^{-5}m^3}{0.0125\,m^2} = 0.00318\,m$$

$$Bi = \frac{(30\ W/m^2K)(0.00318m)}{(13.4\ W/mK)} = 0.007$$

$$\boxed{L_c = 0.00318\,m \quad Bi = 0.007}$$ lumped cap. approx is valid

Lc_2?

Bi_2?

Michael Yanes

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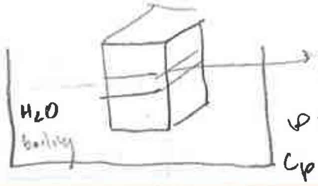

$\rho = 2200\, kg/m^3$

$c_p = 700\, J/kg\cdot K$

0.01 m

$T_i = 30°C$

$T_s = 100°C$

$T(x,t) = T(0.01, 120s) = 65°C$

Problem 5.70

$K = ?$

Equation 5.60 →

$$\frac{T(x,t) - T_s}{T_i - T_s} = erf\left[\frac{x}{2\sqrt{\alpha t}}\right] = \frac{65 - 100}{30 - 100} =$$

$$erf\left[\frac{x}{2\sqrt{\alpha t}}\right] = 0.5 \Rightarrow \frac{0.01}{2\sqrt{\alpha \times 120}} = 0.477 \Leftarrow \text{erf inverse calculator}$$

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$$\sqrt{\alpha \times 120} = \frac{0.01}{2 \times 0.477} \Rightarrow \alpha = \frac{\left(\frac{0.01}{2\times 0.477}\right)^2}{120} = 9.1563 \times 10^{-7}\, m^2/s$$

$$\alpha = \frac{k}{\rho c_p} \Rightarrow K = \alpha \rho c_p = (9.1563 \times 10^{-7})(2200)(700)$$

$$\boxed{K = 1.41\, W/mK}$$

Problem 6.3

$Pr = c_p \cdot \nu / k = 0.7$ $T_\infty = 400K$ $\frac{u_\infty}{\nu} = 5000\, m^{-1}$

$T_s = 300\,K$

$$\frac{T - T_s}{T_\infty - T_s} = 1 - exp\left(-Pr\frac{u_\infty y}{\nu}\right)$$

$q'' = ?$

$T_f = \frac{T_s + T_\infty}{2} = 350K$

$K_f = 0.0296\, W/mK$

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$$T = \left[1 - exp\left(-Pr\frac{u_\infty y}{\nu}\right)\right](T_\infty - T_s) + T_s$$

$$\frac{\partial T}{\partial y} = \left(Pr\frac{u_\infty}{\nu}\right)\left(exp\left(-Pr\frac{u_\infty y}{\nu}\right)\right)(T_\infty - T_s)$$

set $y = 0$

$$\frac{\partial T}{\partial y} = Pr\frac{u_\infty}{\nu}(T_\infty - T_s)$$

$$q_s'' = -K_f \frac{\partial T}{\partial y}\Big|_{y=0} = -K_f Pr \frac{u_\infty}{\nu}(T_\infty - T_s)$$

0.0263

$$q_s'' = -\underset{W/mK}{(0.0296)}\underset{\text{dimensionless}}{(0.7)}\underset{m^{-1}}{(5000)}\underset{K}{(100)}$$

$$\boxed{q_s'' = -10\,360\, W/m^2}$$

$-9205\, W/m^2$